Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
SLOPE FORMULA	The **slope formula** is used to find the slope between two points (x_1, y_1) and (x_2, y_2).
	Formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$
	It is important to remember to SIMPLIFY your answer!
EXAMPLES	**Directions:** Find the slope between each pair of points.

1. (1, 1) and (4, 3)

$m = \frac{3-1}{4-1} = \boxed{\frac{2}{3}}$

2. (-2, 4) and (10, -2)

$m = \frac{-2-4}{10+2} = \frac{-6}{12} = \boxed{-\frac{1}{2}}$

3. (-4, 5) and (-8, -5)

$m = \frac{-5-5}{-8+4} = \frac{-10}{-4} = \boxed{\frac{5}{2}}$

4. (10, 0) and (-2, 4)

$m = \frac{4-0}{-2-10} = \frac{4}{-12} = \boxed{-\frac{1}{3}}$

5. (5, 9) and (3, 9)

$m = \frac{9-9}{3-5} = \frac{0}{-2} = \boxed{0}$

6. (-7, 8) and (-7, 5)

$m = \frac{5-8}{-7+7} = \frac{-3}{0} = \boxed{\text{undef.}}$

7. (-1, 9) and (2, 3)

$m = \frac{3-9}{2+1} = \frac{-6}{3} = \boxed{-2}$

8. (-4, 13) and (6, -2)

$m = \frac{-2-13}{6+4} = \frac{-15}{10} = \boxed{-\frac{3}{2}}$

9. (5, 6) and (6, 5)

$m = \frac{5-6}{6-5} = \frac{-1}{1} = \boxed{-1}$

10. (9, -4) and (1, -4)

$m = \frac{-4+4}{1-9} = \frac{0}{-8} = \boxed{0}$

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11. (5, -9) and (3, -2)

$m = \frac{-2+9}{3-5} = \boxed{\frac{7}{-2}}$

12. (4, 6) and (4, 8)

$m = \frac{8-6}{4-4} = \frac{2}{0} = \boxed{\text{undef.}}$

GOING BACKWARDS

Sometimes you must complete the ordered pairs using a given slope.

Example: If the slope of the line passing through the points (-5, 6) and $(5, y)$ is $-\frac{4}{5}$, find y.

$\frac{-4}{5} = \frac{y-6}{5+5}$ $\quad \frac{-4}{5} = \frac{y-6}{10}$ $\quad -40 = 5y - 30$

$-10 = 5y$

$\boxed{-2 = y}$

MORE EXAMPLES

Directions: Find the missing value using the given slope.

13. (-3, -2) and $(x, 6)$; $m = 2$

$2 = \frac{6+2}{x+3}$ $\quad 2x+6 = 8$

$2x = 2$

$\boxed{x = 1}$

14. (0, -4) and $(x, -7)$; $m = \frac{3}{2}$

$\frac{3}{2} = \frac{-7+4}{x-0}$ $\quad \frac{3}{2} = \frac{-3}{x}$

$3x = -6$

$\boxed{x = -2}$

15. (-3, -4) and $(-5, y)$; $m = -\frac{9}{2}$

$\frac{-9}{2} = \frac{y+4}{-5+3}$ $\quad \frac{-9}{2} = \frac{y+4}{-2}$

$2y + 8 = 18$

$2y = 10$

$\boxed{y = 5}$

16. $(x, 2)$ and (6, 3); $m = -\frac{1}{2}$

$-\frac{1}{2} = \frac{3-2}{6-x}$ $\quad -\frac{1}{2} = \frac{1}{6-x}$

$-6 + x = 2$

$\boxed{x = 8}$

17. $(-3, y)$ and (1, -7); $m = -4$

$-4 = \frac{-7-y}{1+3}$ $\quad -7 - y = -16$

$-y = -9$

$\boxed{y = 9}$

18. $(x, 7)$ and (11, 8); $m = -\frac{1}{5}$

$-\frac{1}{5} = \frac{8-7}{11-x}$ $\quad -\frac{1}{5} = \frac{1}{11-x}$

$5 = -11 + x$

$\boxed{16 = x}$

19. $(4, y)$ and (0, 5); $m = \frac{3}{4}$

$\frac{3}{4} = \frac{5-y}{0-4}$ $\quad \frac{3}{4} = \frac{5-y}{-4}$

$20 - 4y = -12$

$-4y = -32$

$\boxed{y = 8}$

20. $(-3, y)$ and (9, -2); $m = \frac{1}{3}$

$\frac{1}{3} = \frac{-2-y}{9+3}$ $\quad \frac{1}{3} = \frac{-2-y}{12}$

$12 = -6 - 3y$

$18 = -3y$

$\boxed{-6 = y}$

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